

## **Data Structures - Subjective Test (20 Marks)**

**Answer the following questions:**

1. 1. Explain the difference between a singly linked list and a doubly linked list. (4 Marks)
2. 2. Describe the concept of dynamic memory allocation in data structures. (4 Marks)
3. 3. What is a priority queue? How is it implemented using a heap? (4 Marks)
4. 4. Explain the concept of binary search and its time complexity. (4 Marks)
5. 5. Discuss the applications of graph data structures in real life. (4 Marks)